Aayush Gupta
SRN: PES2UG24CS018

Lab 3: Architecture Justification

Architecture Selection

We chose Microservices Architecture for the Remote Patient Vitals Alert and Monitoring System.

Reason 1: The parts of this system carry very different loads

NFR-001 requires the telemetry gateway to sustain 500 concurrent streams, while threshold configuration is used rarely, perhaps once when a patient is discharged. As separate services, only Telemetry Ingestion and Vitals Evaluation need to be replicated at peak load. In a layered or client-server design the whole application is one deployable unit, so scaling for ingestion means paying to replicate the configuration and alerting code that nobody is calling.

Reason 2: The alert path must survive failures elsewhere

NFR-001 sets a 99.99% uptime target, and for a post operative patient a missed alert is a clinical risk rather than an inconvenience. With independent services, a crash in the Threshold Configuration Service does not stop Vitals Evaluation from checking readings against limits already held, and alerts still reach caregivers. In a layered design a failure in a shared layer takes the alert path down with it, and in a client-server design the single server is one point of failure for the whole system.

Security Advantage

Each service authenticates to the Patient Data Store separately and is granted access only to the records it needs, which is how NFR-002 restricts a patient record to the caregivers on that patient's contact list. The Alert Escalation Service reads the caregiver contact list but never holds the full vitals history, so if the caregiver facing service is compromised the attacker does not inherit the whole record set. All traffic between services runs over TLS, which covers the in transit half of NFR-002.

Performance Benefit

FR-002 gives a 2 second budget from a reading arriving to an anomaly being flagged. Because Telemetry Ingestion and Vitals Evaluation are separate deployables, instances can be added to just those two services as stream count rises. Throughput on the detection path goes up without touching the rest of the system, so the 95th percentile latency stays under 2 seconds at 500 streams without over provisioning services that see almost no traffic.

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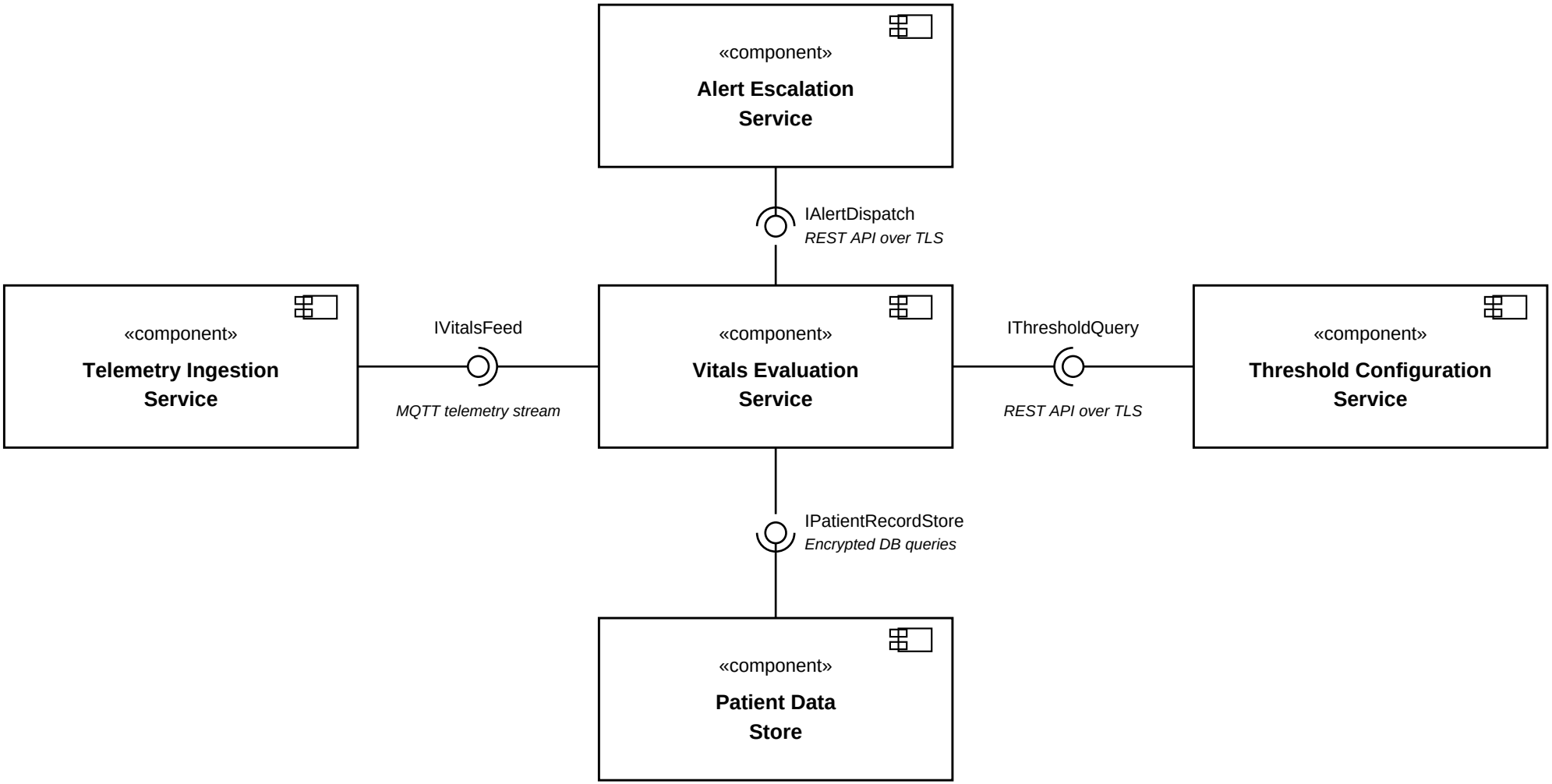
Lab 3: Component Diagram

Remote Patient Vitals Alert and Monitoring System | Microservices Architecture